Lecture 38
2nd Order Transients – 4 of 4

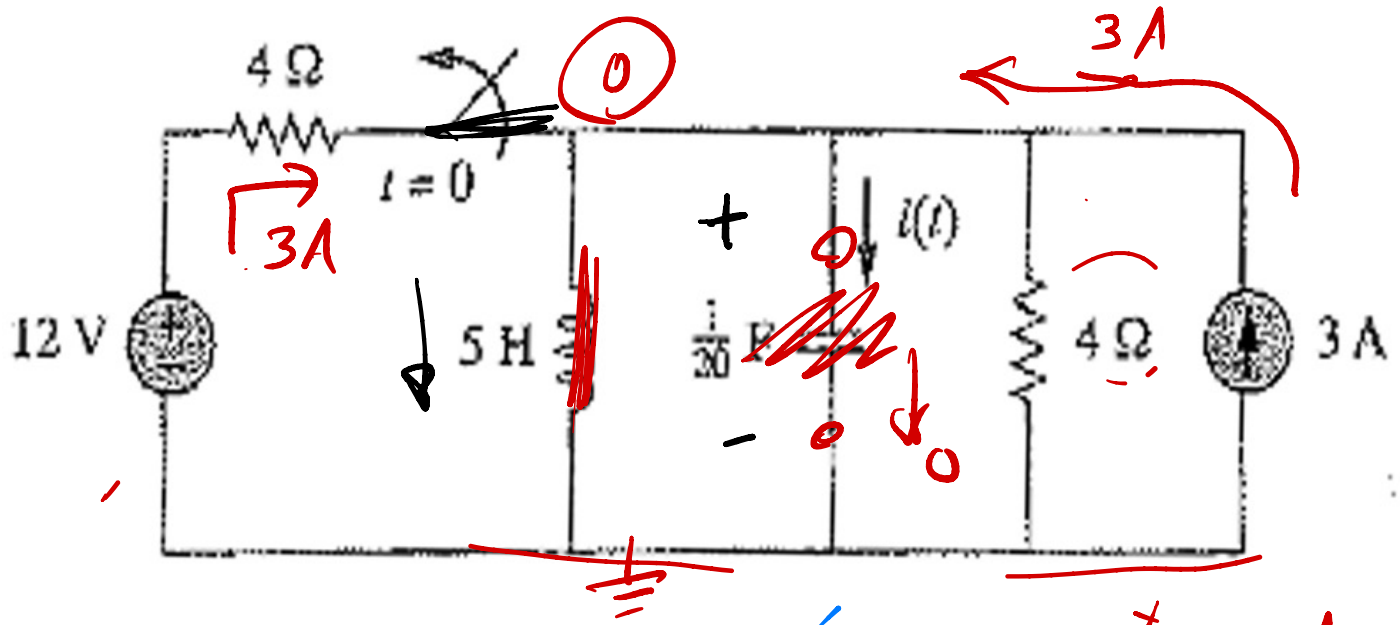
examples

Beyond $i_L(t)$ or $v_C(t)$

1. Identify type (series/parallel) and values of R,L,C
2. Root characteristic equation, to find form
3. Find $i_L(0)$ and $v_C(0)$
4. For variable of interest, find $x(0), x'(0)$, and $x(\infty)$
   - Note that the derivative might be hard to find; alternatively, could solve for $i_L(t)$ or $v_C(t)$ and then "propagate" that answer to the desired variable
5. Assemble answer

$i_C = C \frac{dv}{dt}$

Example: find the capacitor current for $t > 0$. Hint, try v_c instead.



$s^2 + 5s + 4 = 0$

$(s+4)(s+1)$

$s_1 = -4$

$= -1$

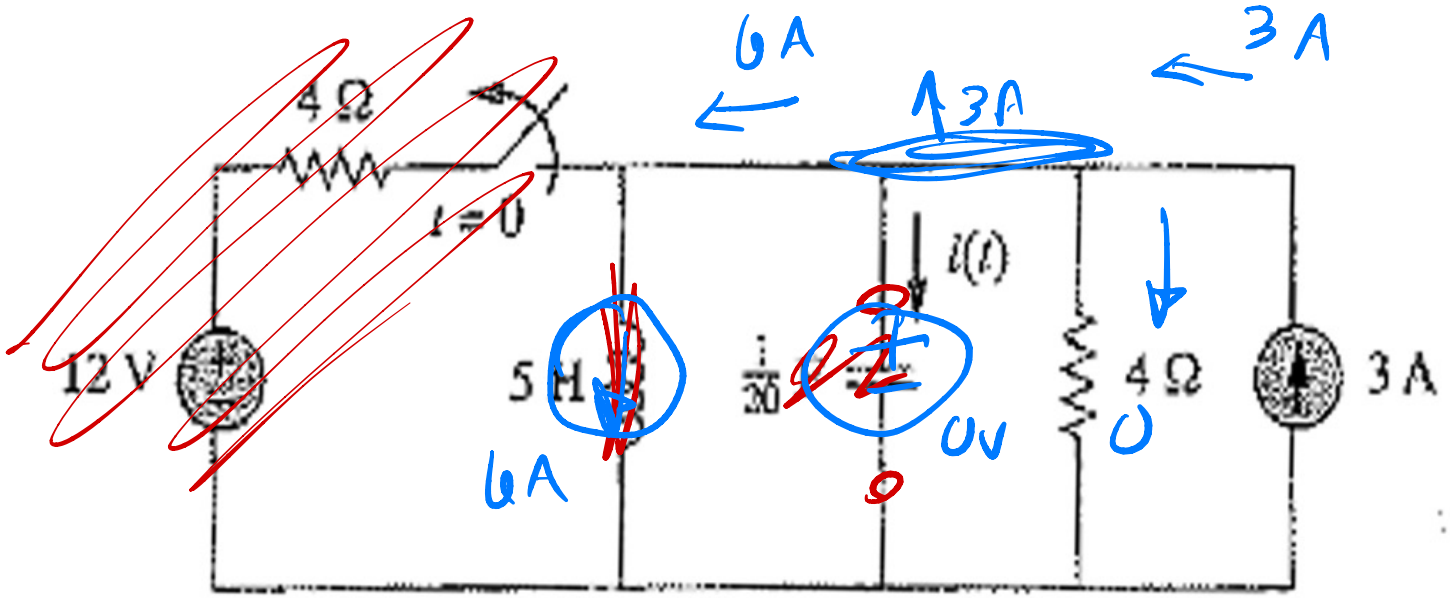
$t<0$

$\cancel{i_\infty} + A_1 e^{-t} + A_2 e^{-4t}$

0

$i_L(t) = 6\ A \quad t \le 0$

$v_c(t) = 0\ V \quad t \le 0$



$i_\infty = 0$

$i_L(0) = 6\ A$

$v_c(0) = 0\ V$

$s_1 = -1$

$s_2 = -4$

i_c

$v_c(t) = v_\infty + A_1 e^{-t} + A_2 e^{-4t}\ V$

$\underline{0}$

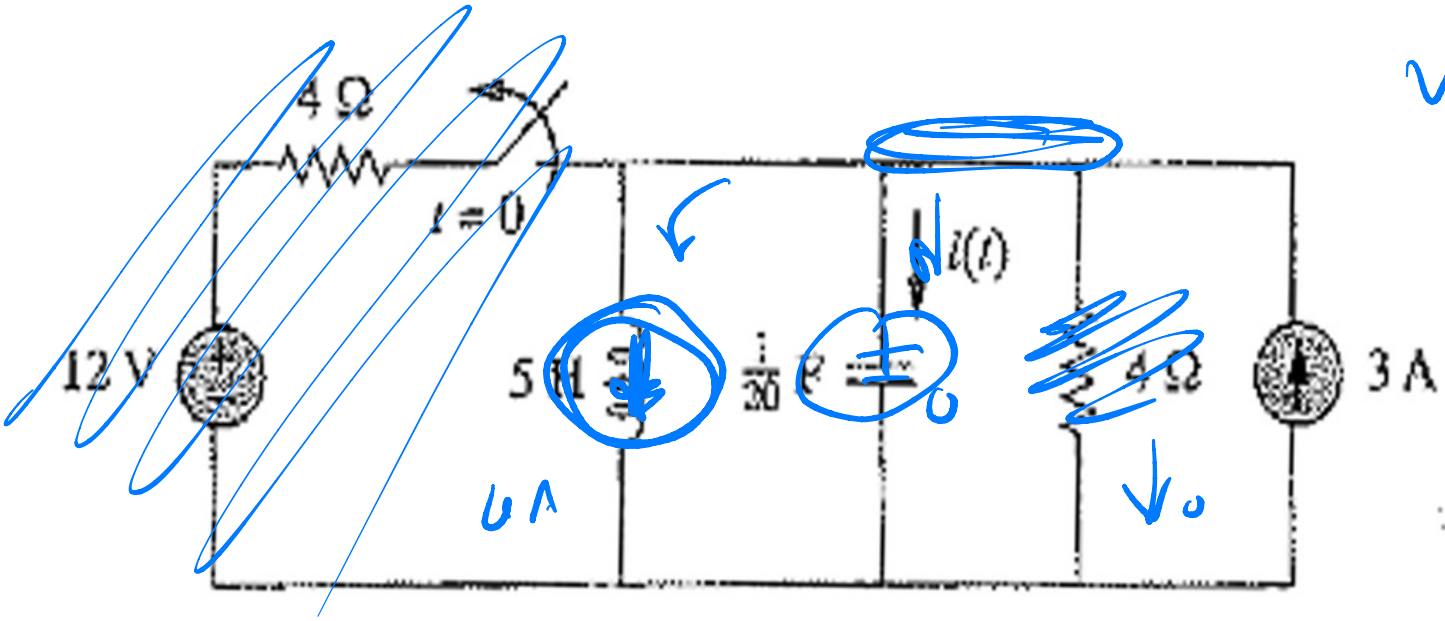
$A_1 = \dfrac{v_c'(0)}{3}$ $\quad A_2 = -A_1$

$A_1 = -30,\ A_2 = +30$

$i_c = C\dfrac{dv}{dt} = C \cdot v'(0)$

$i_c(0) = C\,v_c'(0) = -3\ A$ $\quad v_c'(0) = \dfrac{-3}{1/30} = -90$

$i(t) = e^{-t} - e^{-4t}\ A$



$v_c(t)$

$v_c'(0)$

$i_c = C \frac{dv}{dt}$

$= C\, v'(t)$

$6 + i_c + 0 - 3 = 0$

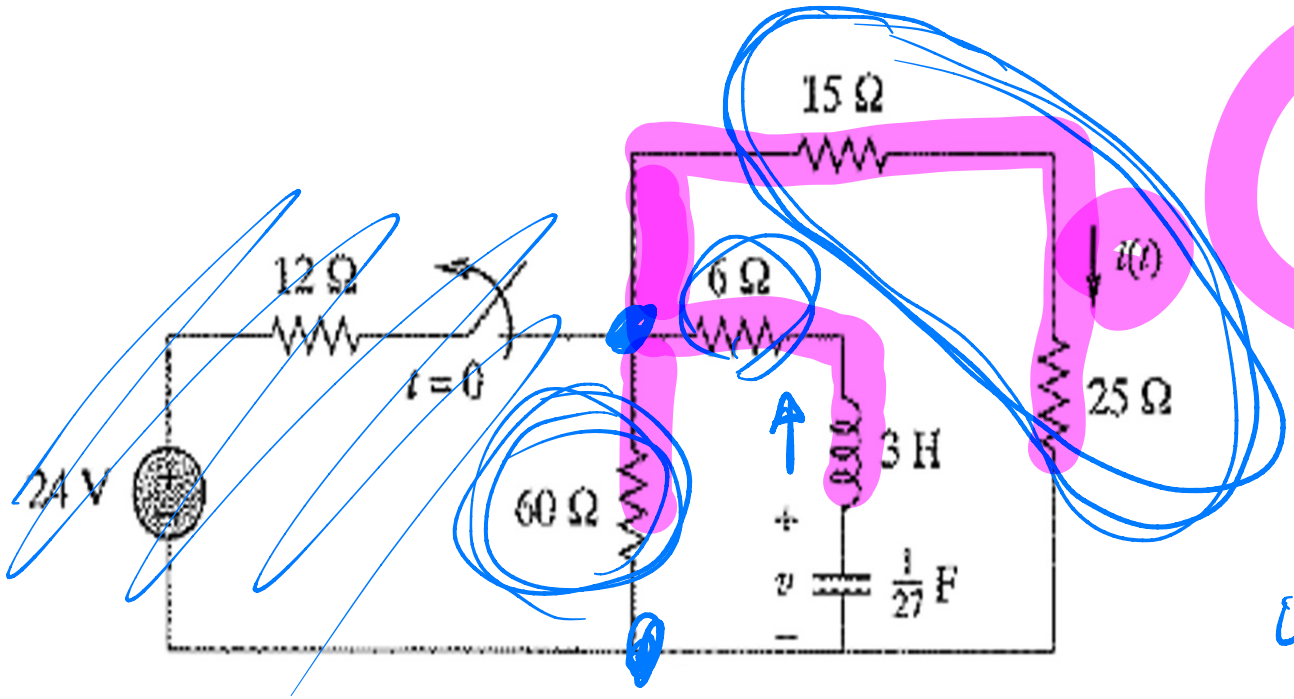
$v'(0) = \frac{1}{C}\, i_c(0)$

$i_c = -3$

$v_c(t) = -e^{-t} + e^{-4t}$

$i(t) = e^{-t} - e^{-4t} A$

Example: Find the 25 ohm resistor's current for $t > 0$



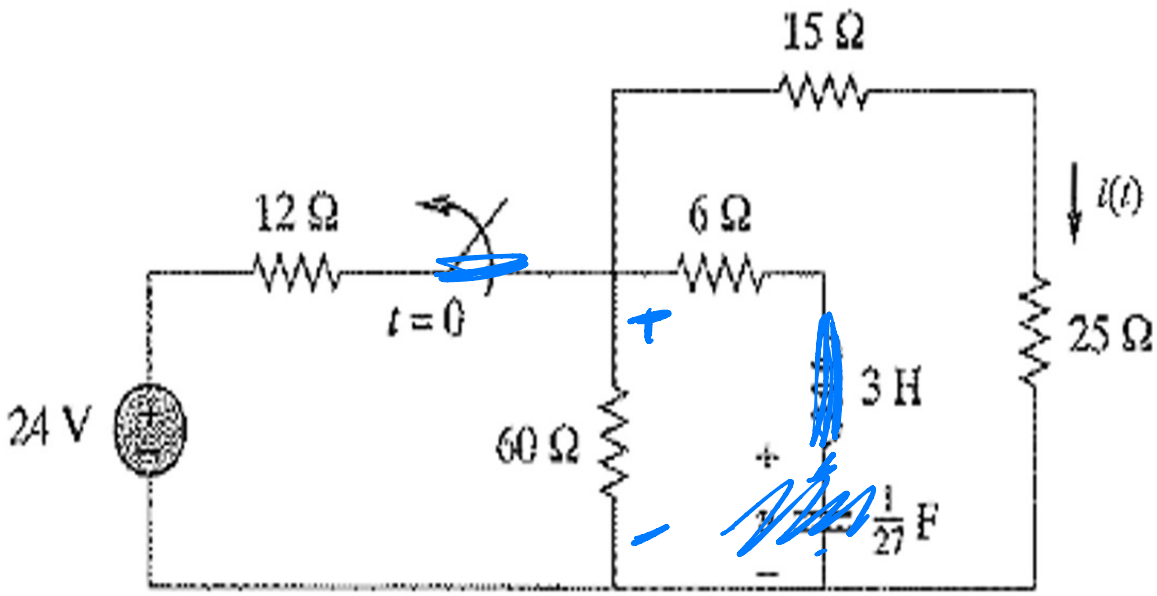
Series RLC

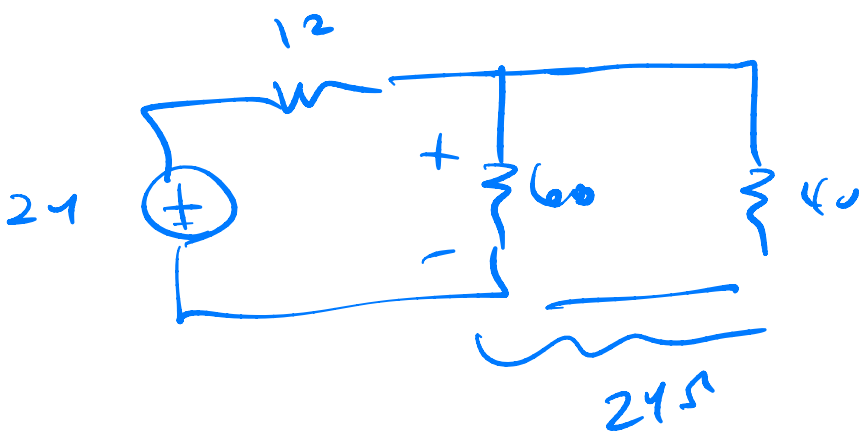
$R = 30\ \Omega$

$i(t) = \frac{60}{40+60} \cdot i_L(t)$

$s^2 + 10s + 9 = 0$

$s_1 = -1 \quad s_2 = -9$





$i_{(0)} = 0$

$i_L(0) = 0$

$v_c(0) = 16\ V$

$i_L'(0)$

$v = L\ di_L(t)$

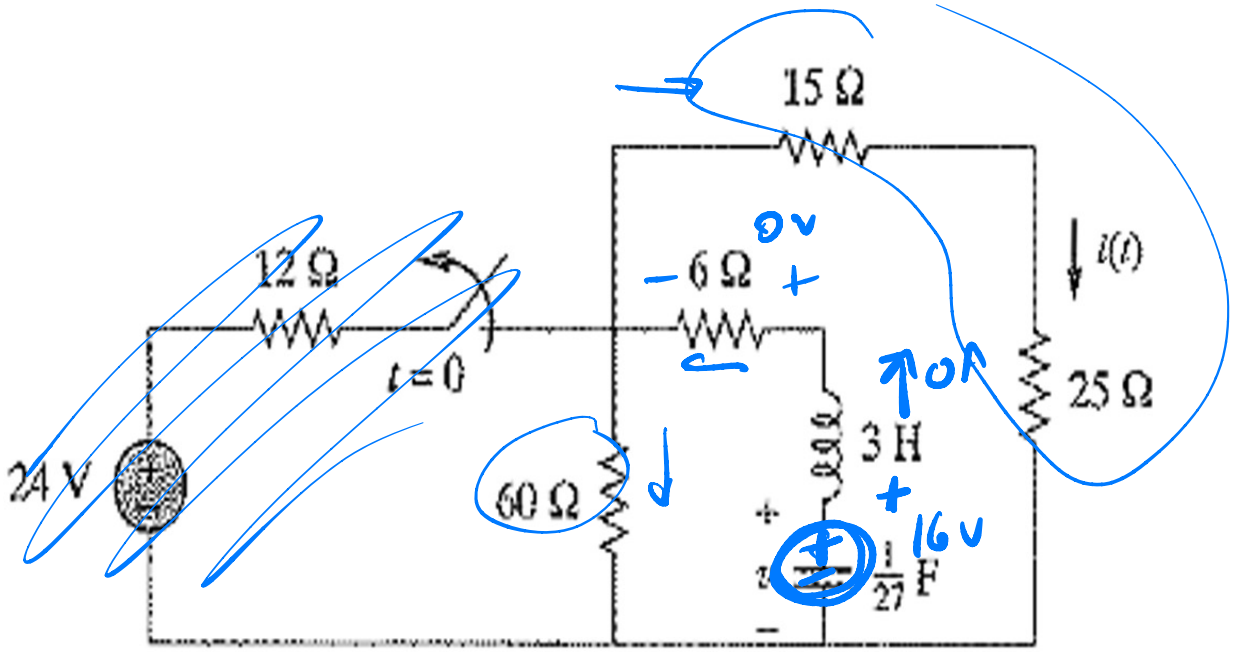
$V_L(0) = L \cdot i_L'(0)$

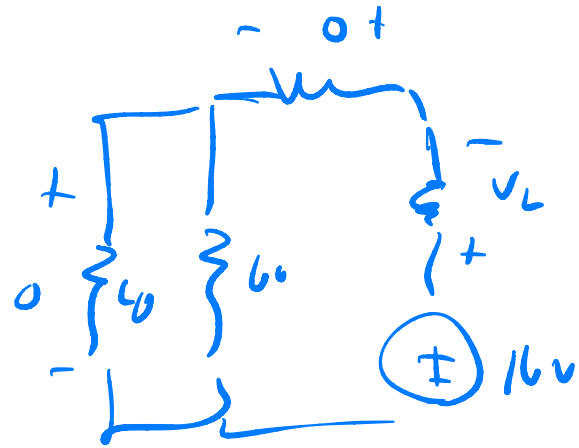
$16 = 3\ i_L'(0)$

$i_L'(0) = \frac{16}{3}$

$$v(t) = 27e^{-t} - 3e^{-9t}\ V$$

$$i_{25}(t) = 0.6(e^{-t} - e^{-9t})\ A$$





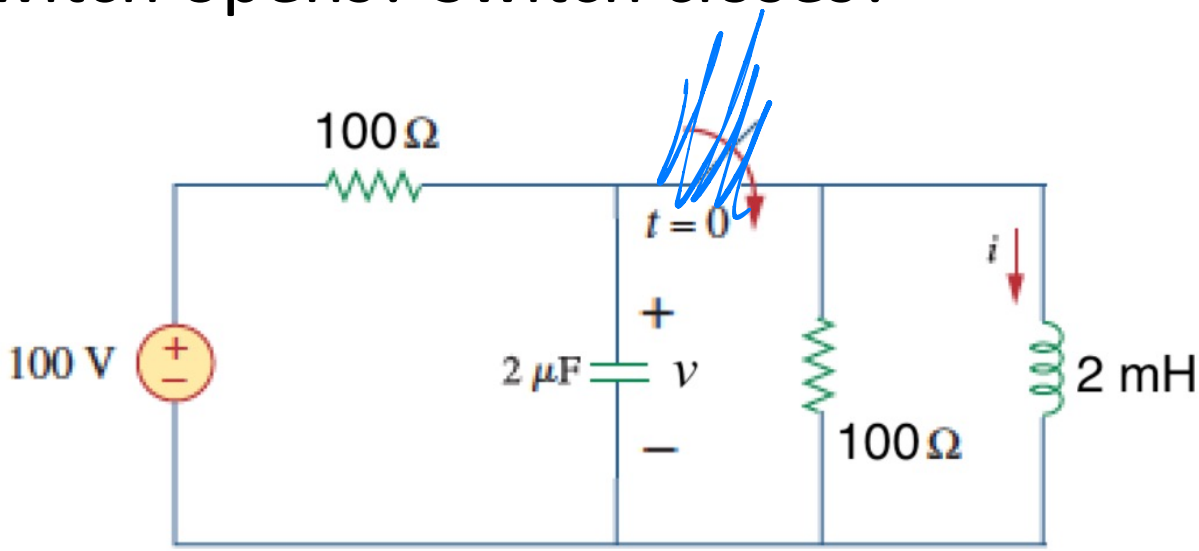
$v_L(0) = 16$

$$v(t) = 27e^{-t} - 3e^{-9t}\ V$$

$$i_{25}(t) = 0.6(e^{-t} - e^{-9t})\ A$$

Other Situations

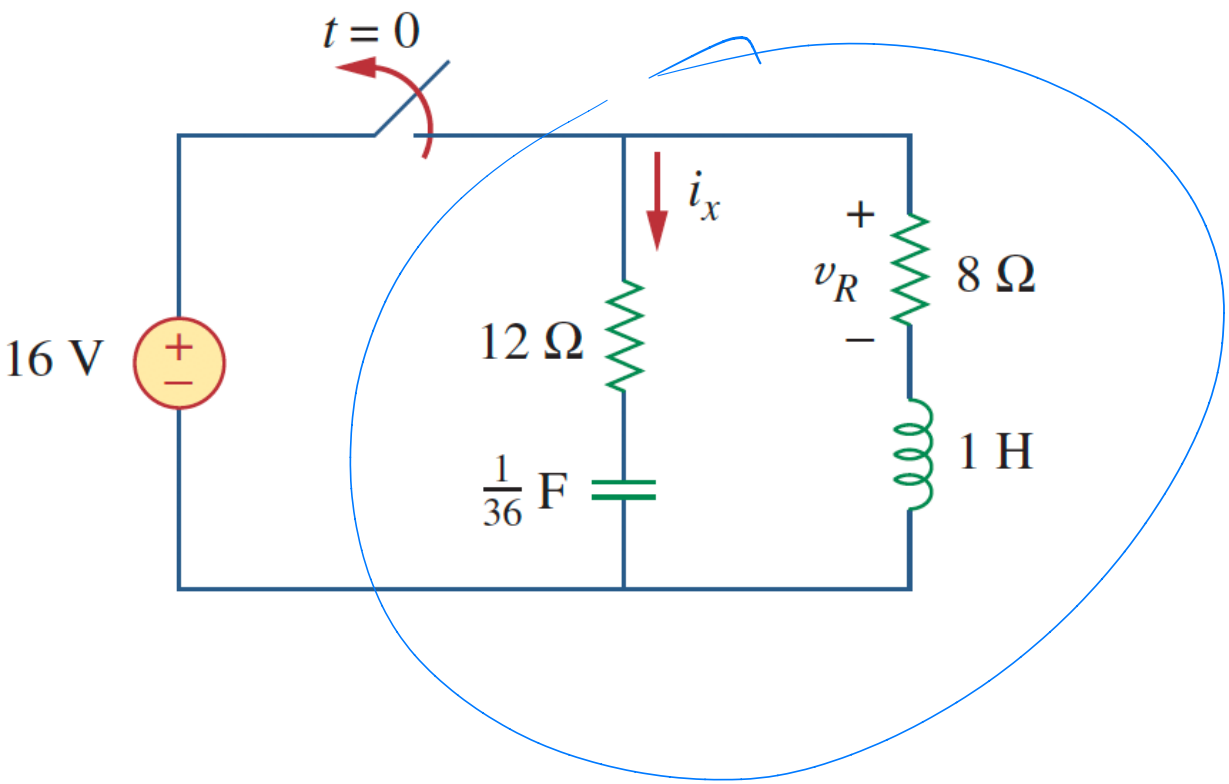
Switch opens? Switch closes?



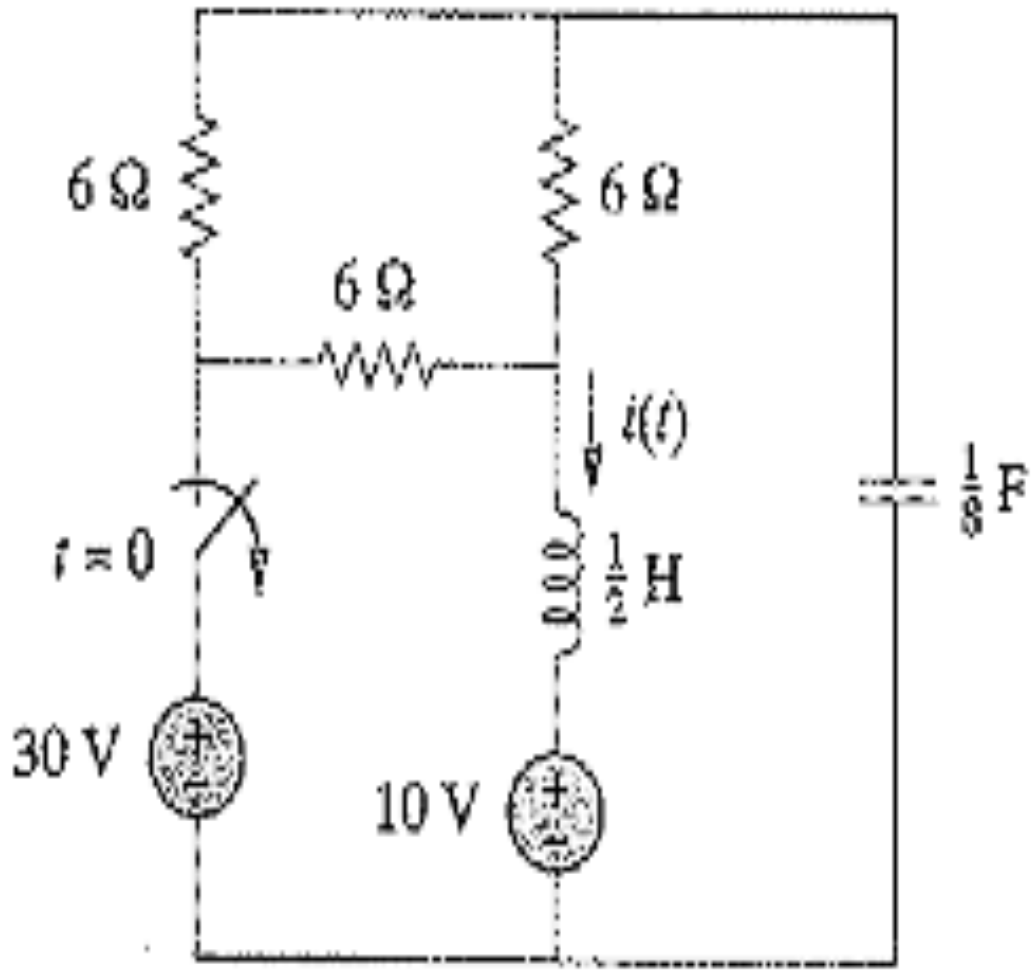
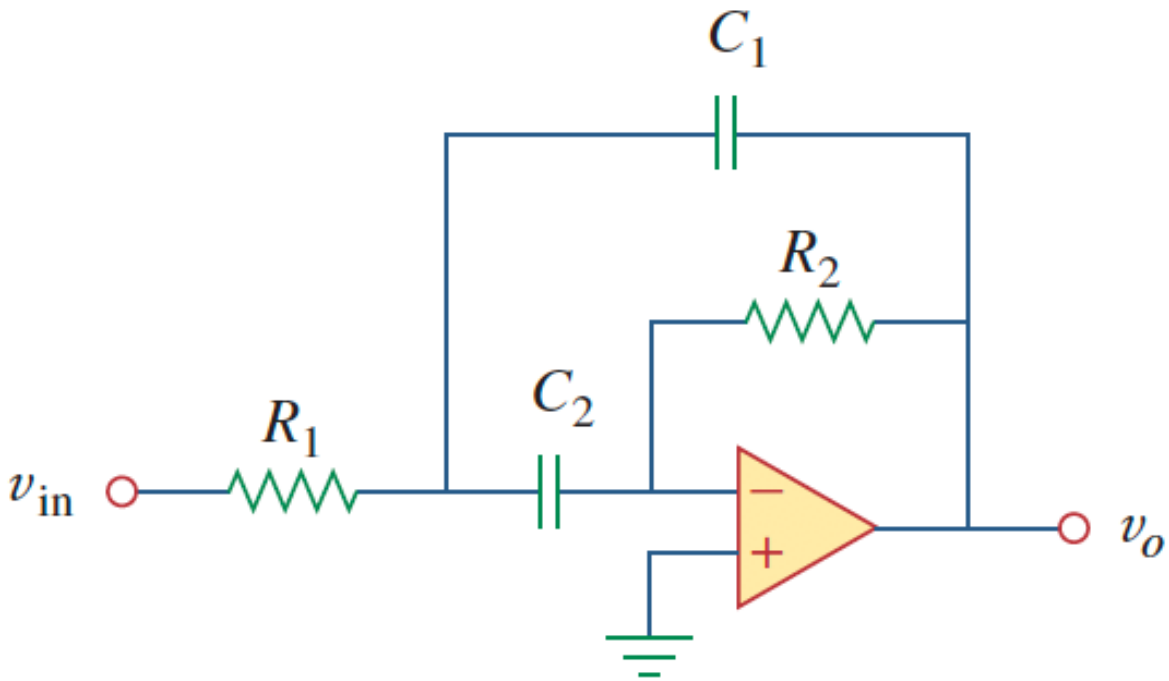
closes

R = 50 Ω

- Switch opens? Switch closes?



- Switch opens? Switch closes?



What would you do here if the input was a unit step?